

Decent Coaching Classes

Std. IX (CBSE) — Mind Your Math - I

Answer Key — Ch 2, Ch 4 & Ch 7 (Ganita Manjari)

Section A

(1) (b) 1

(2) (a) 3

(3) (c) $a^2 + 2ab + b^2$

(4) (c) $1/2$

(5) (a) 9975 [$105 \times 95 = (100+5)(100-5) = 100^2 - 5^2 = 10000 - 25 = 9975$]

Section B

(1) $3x + 12 = 0 \Rightarrow x = -4$

(2) $p(2) = 4(2) - 8 = 8 - 8 = 0$. Since $p(2) = 0$, $x = 2$ is a zero of $p(x)$.

(3) $(2x + 3y)^2 = 4x^2 + 12xy + 9y^2$

(4) Numbers greater than 4 on a die: {5, 6}. $P(\text{number} > 4) = 2/6 = 1/3$

(5) $998^2 = (1000 - 2)^2 = 1000000 - 4000 + 4 = 996004$

Section C

(1) $p(0) = 5(0) - 2 = -2$; $p(1) = 5(1) - 2 = 3$; $p(-1) = 5(-1) - 2 = -7$. Since $p(1) = 3 \neq 0$, $x = 1$ is NOT a zero of $p(x)$.

(2) $4x^2 - 25y^2 = (2x)^2 - (5y)^2 = (2x - 5y)(2x + 5y)$

(3) Outcomes giving sum 8 (out of 36): (2,6), (3,5), (4,4), (5,3), (6,2) $\rightarrow$ 5 outcomes. $P(\text{sum} = 8) = 5/36$

Section D

(1)(a) LHS: $(2+3)^3 = 5^3 = 125$. RHS: $2^3 + 3(2^2)(3) + 3(2)(3^2) + 3^3 = 8 + 36 + 54 + 27 = 125$. Since LHS = RHS, the identity is verified.

(1)(b) Total balls = $3 + 4 + 5 = 12$. (i) $P(\text{red}) = 3/12 = 1/4$ (ii) $P(\text{not green}) = 1 - 5/12 = 7/12$